

NYC Taxi Trip Duration — Regression Report

23CSE301 Machine Learning | Review 1 | Regression track

Data file: /Users/bettim/Documents/Semester 5/Machine Learning/Capstone Evaluation/data/nyc_taxi_trip_duration.csv

Raw rows loaded: 774,399

Rows after cleaning: 767,882

Rows in modelling table: 767,882

Training rows: 614,305 Evaluation rows: 50,000

FAST_DEV = False

Best model by test R^2 : Random Forest Regressor

Official models trained: 12

Figures exported: 11

Pipeline: audit → EDA → clean → engineer → split/encode/scale

→ 10 models → compare/tune/CV → manual tester.

Encoders and scalers were fit on the training split only.

dropoff_datetime and id were never used as predictors.

Data cleaning — rows removed at each step

step	rows_before	rows_after	rows_removed	pct_removed	note
0. loaded	774399	774399	0	0.0000	raw file
1. drop duplicate rows	774399	774399	0	0.0000	exact row duplicates
2. drop leakage / id columns	774399	774399	0	0.0000	id + dropoff_datetime removed (column dropped)
3. NYC bounding box	774399	773489	910	0.1175	$[-74.30,-73.70] \times [40.50,40.90]$
4. passenger_count 1–6	773489	773458	31	0.0040	drop 0 and 7+
5. duration 60–7200 s	773458	767882	5576	0.7209	drop cancelled / multi-day trips

Test metrics (same held-out trips for every model)

Rank	Model	Train rows	R2	RMSE	MAE	Fit seconds
1	Random Forest Regressor	614305	0.7957	293.1957	187.1681	43.8561
2	Gradient Boosting Regressor	614305	0.7944	294.1041	188.4971	3.6134
3	Random Forest Regressor (tuned)	614305	0.7939	294.4709	188.6294	42.5277
4	Decision Tree Regressor	614305	0.7551	321.0052	209.6959	5.0859
5	Support Vector Regressor	12000	0.7245	340.4377	215.2715	2.9115
6	Polynomial Regression	614305	0.6985	356.1212	239.9862	4.1037
7	K-Nearest Neighbors Regressor	40000	0.6935	359.0914	246.2539	0.0005
8	Ridge Regression	614305	0.6222	398.6780	274.3572	0.0307
9	Ridge Regression (tuned)	614305	0.6222	398.6780	274.3572	0.0303
10	Linear Regression	614305	0.6222	398.6780	274.3569	0.1637
11	Lasso Regression	614305	0.6219	398.8196	274.5795	3.4296
12	ElasticNet Regression	614305	0.6063	407.0028	283.2807	0.3143
13	Mean baseline	614305	-0.0000	648.6169	463.8200	0.0008

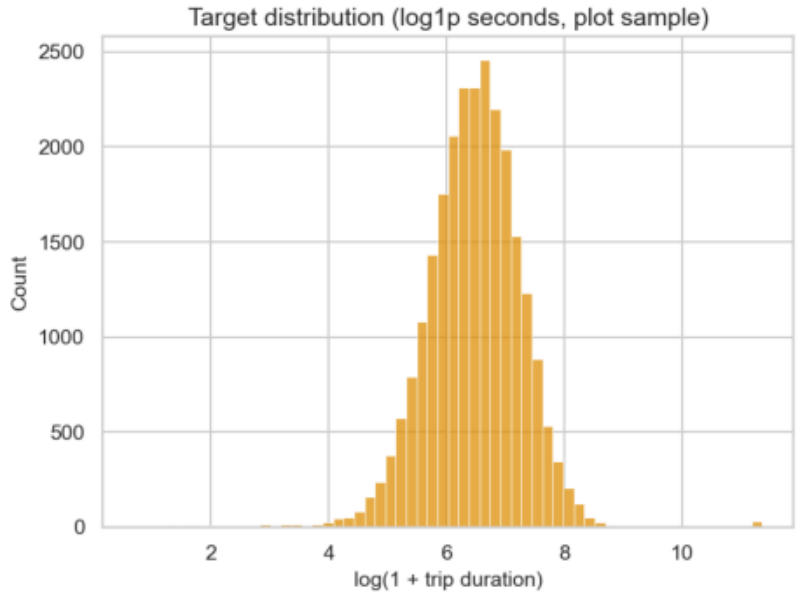
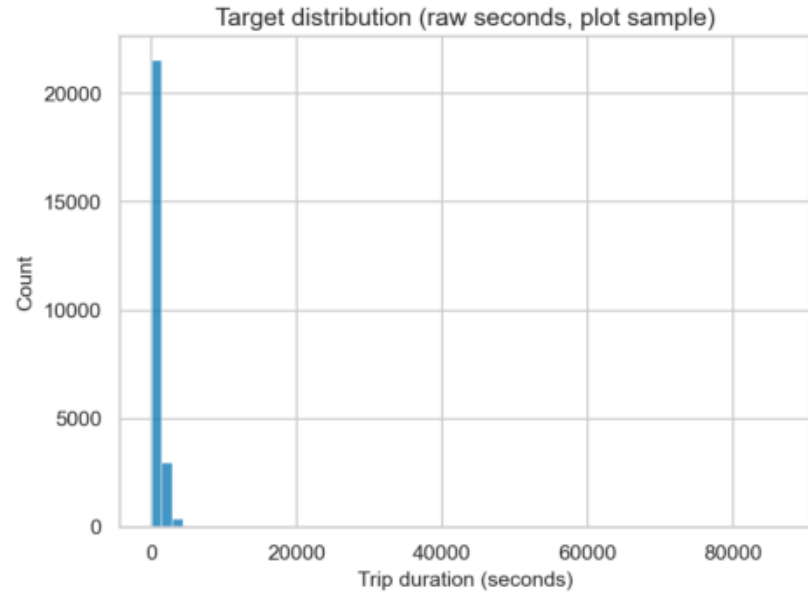
5-fold CV R² for the two best official models

Model	CV folds	CV R2 mean	CV R2 std	Fold scores
Random Forest Regressor	5	0.7924	0.0035	[0.7972 0.7874 0.7947 0.7926 0.7899]
Gradient Boosting Regressor	5	0.7926	0.0035	[0.7971 0.7877 0.7958 0.7921 0.7902]

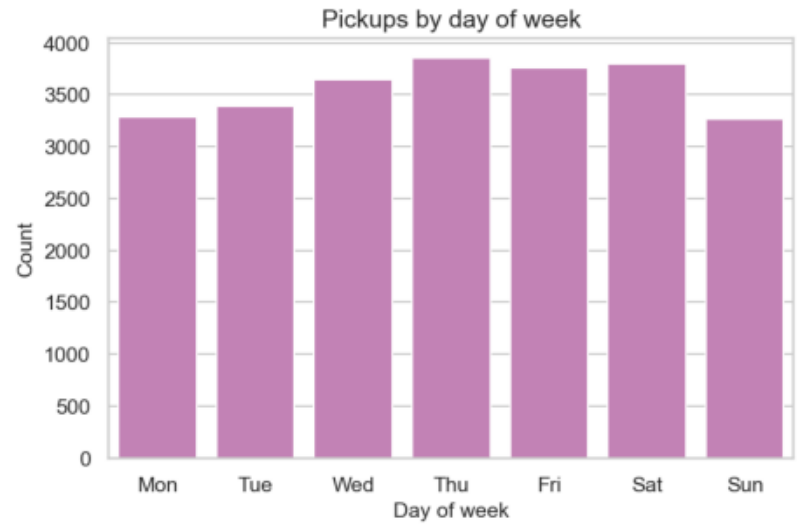
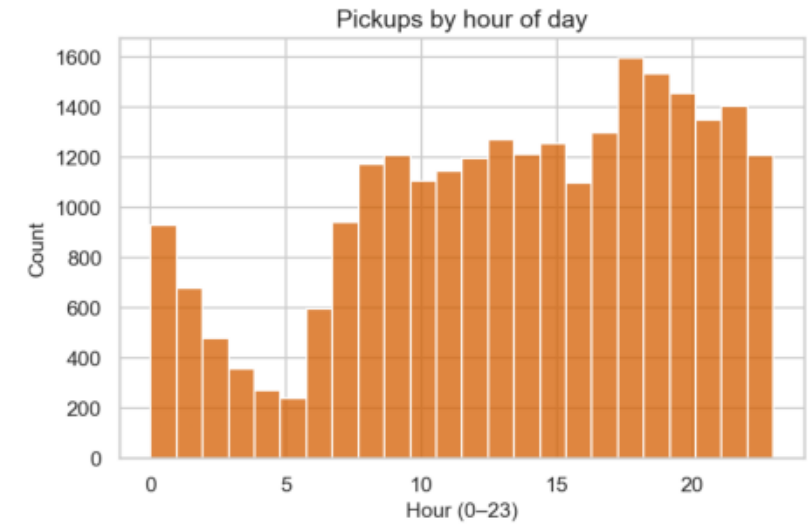
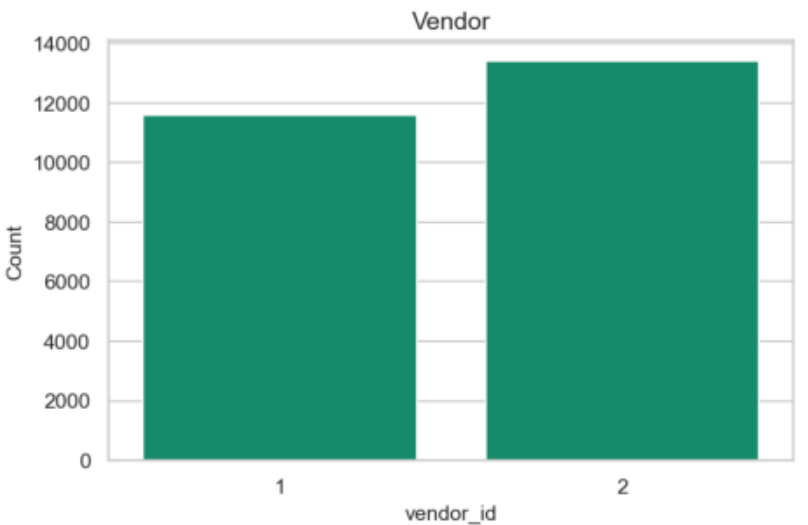
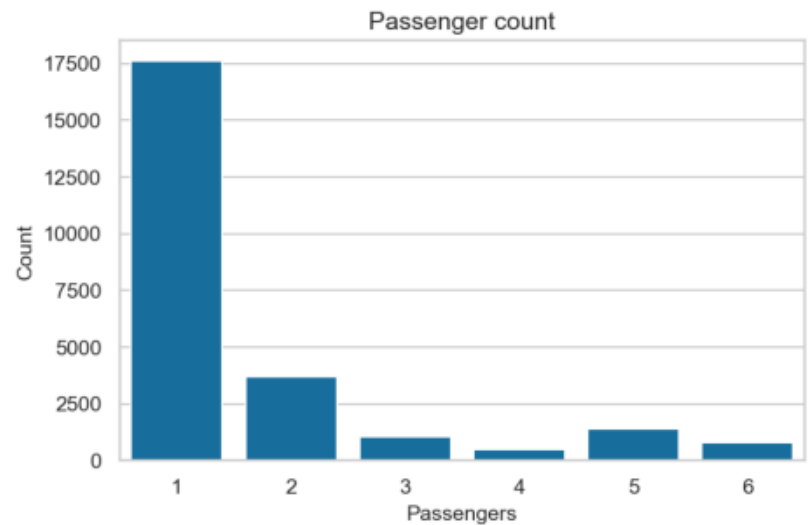
Manual tester — predicted duration for user-edited trips

trip	model_used	predicted_seconds	predicted_minutes	trip_distance_km	pickup_hour	is_rush_hour
1	Random Forest Regressor	715.9631	11.9327	1.5041	17	1
2	Random Forest Regressor	613.0021	10.2167	1.8075	0	0

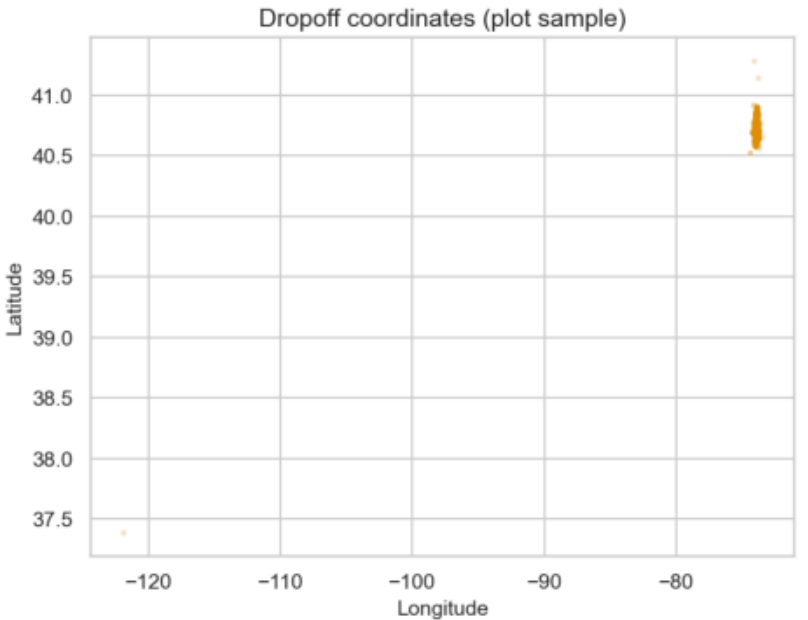
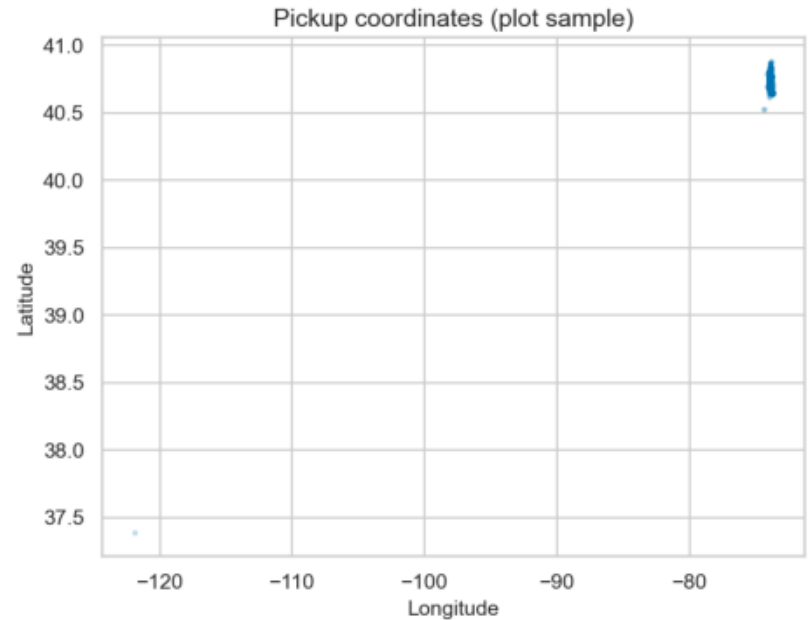
01 target distribution raw seconds plot sample



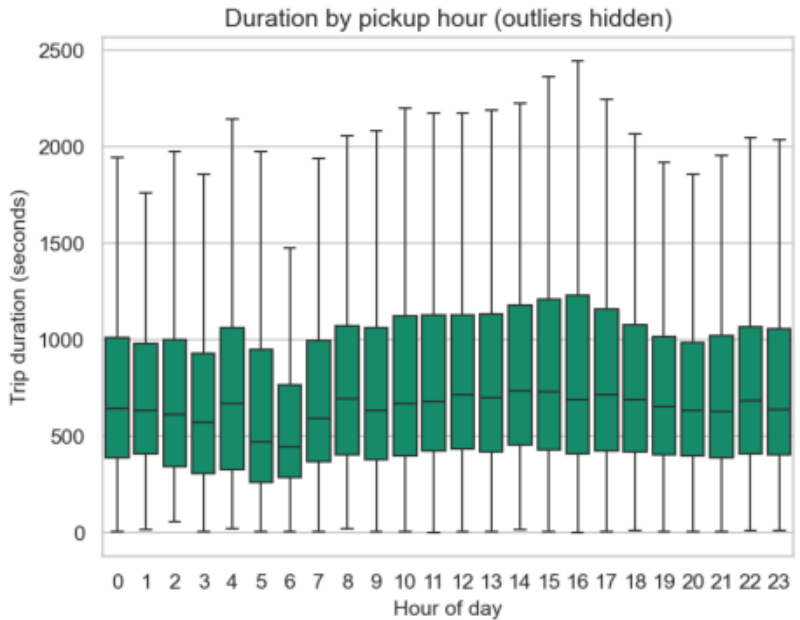
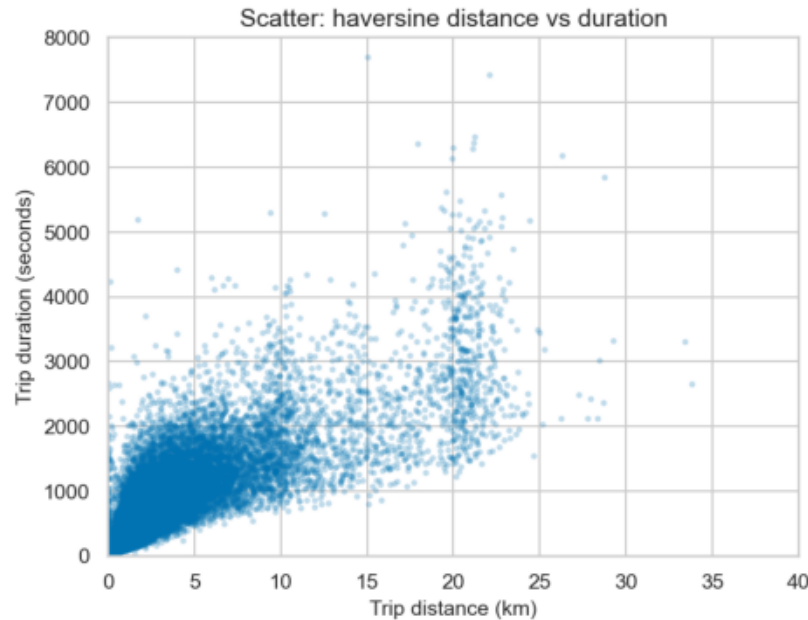
02 passenger count

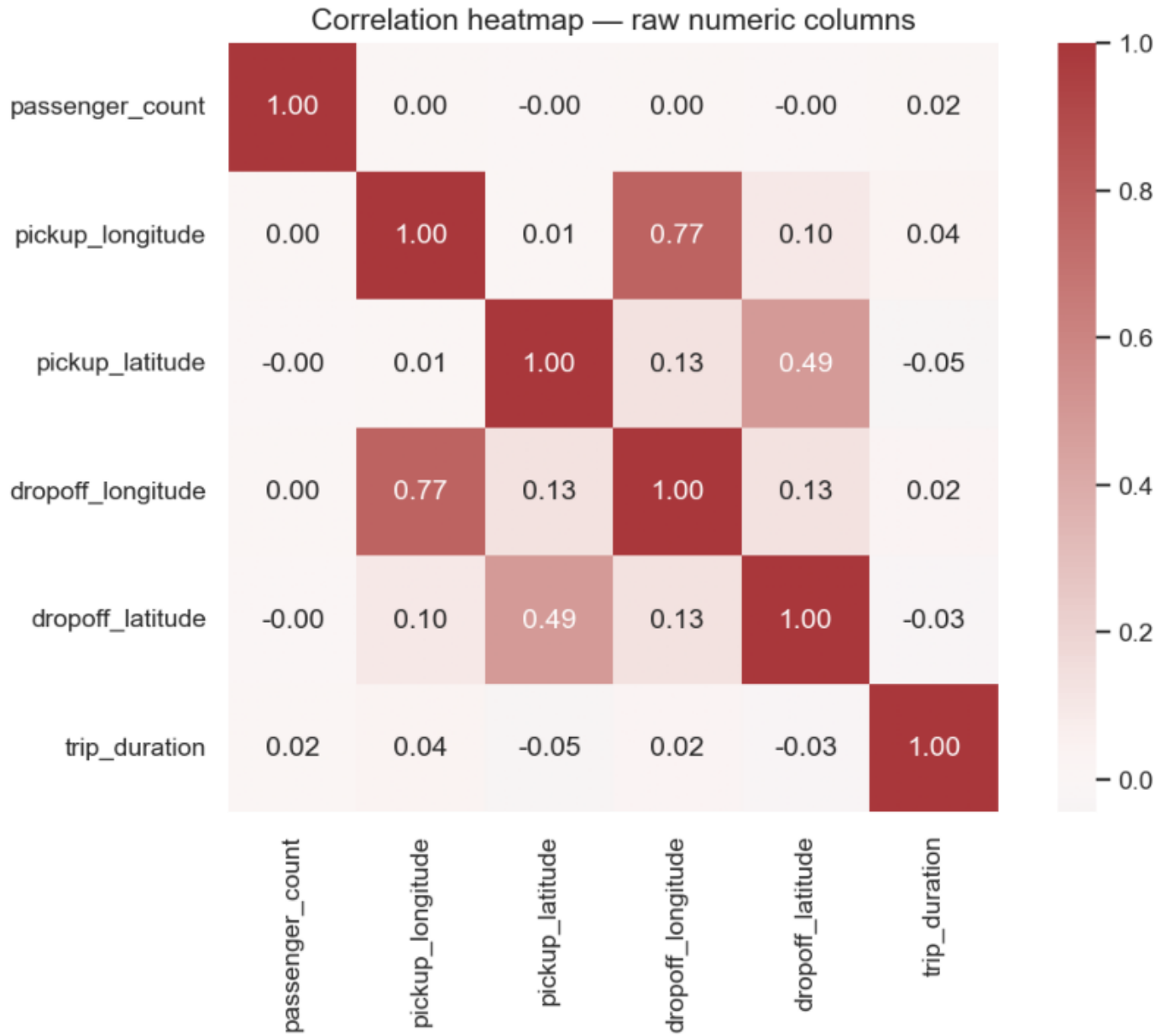


03 pickup coordinates plot sample

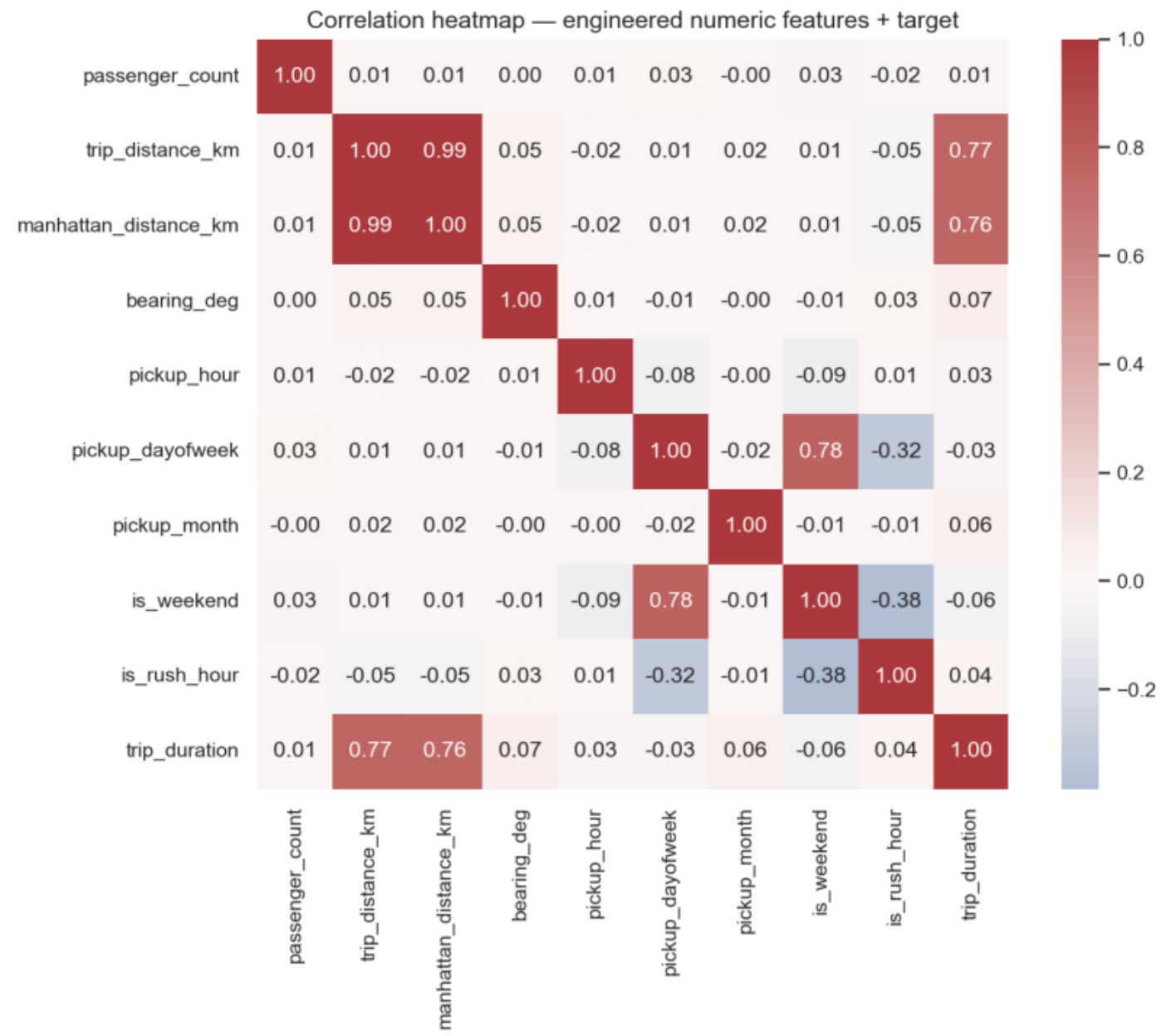


04 scatter haversine distance vs duration

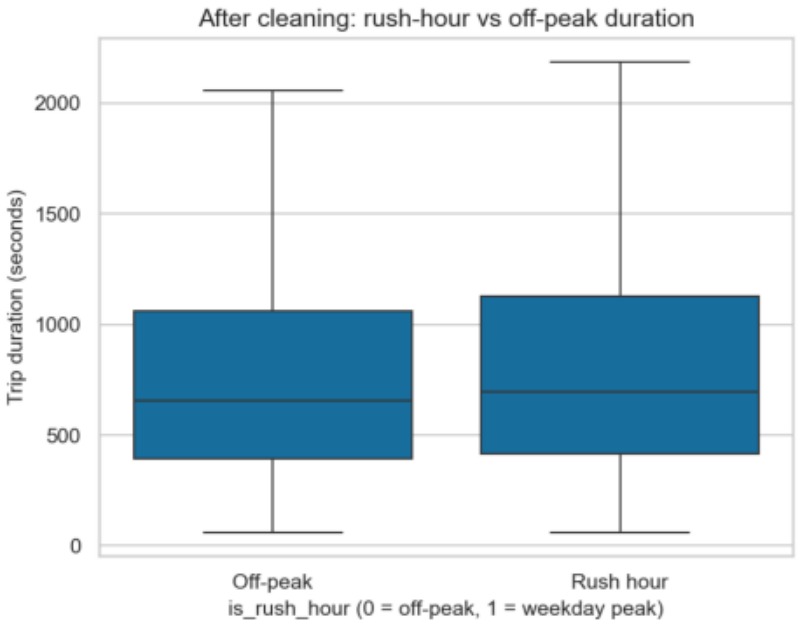
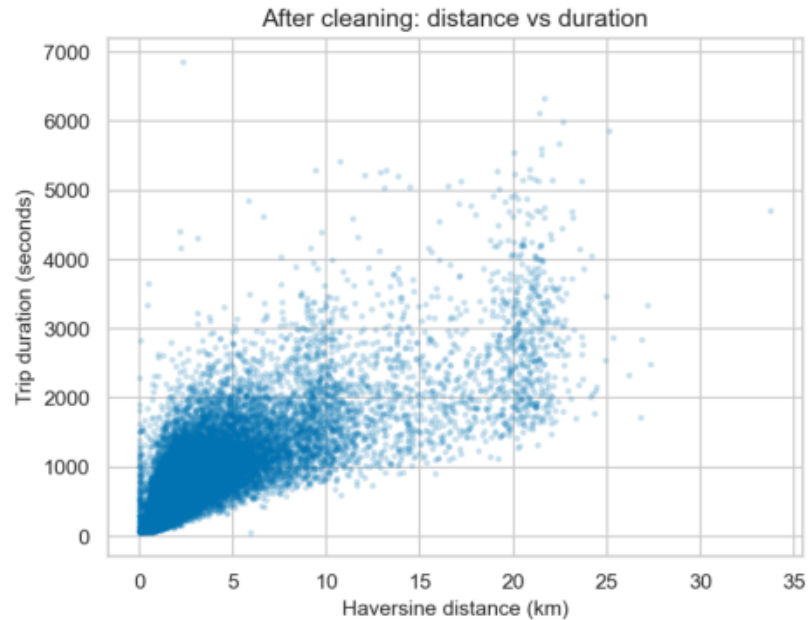


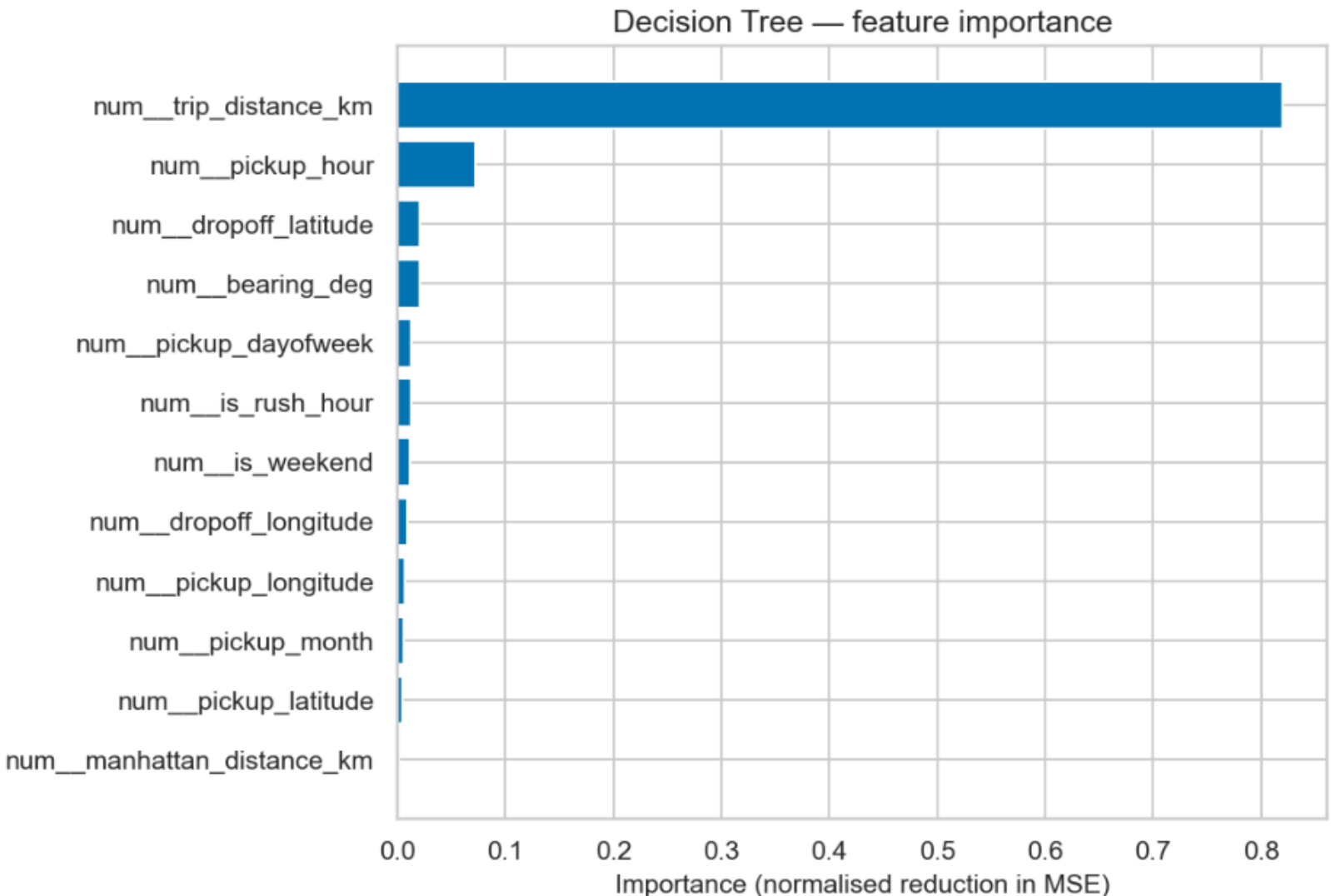


06 correlation heatmap engineered numeric features target

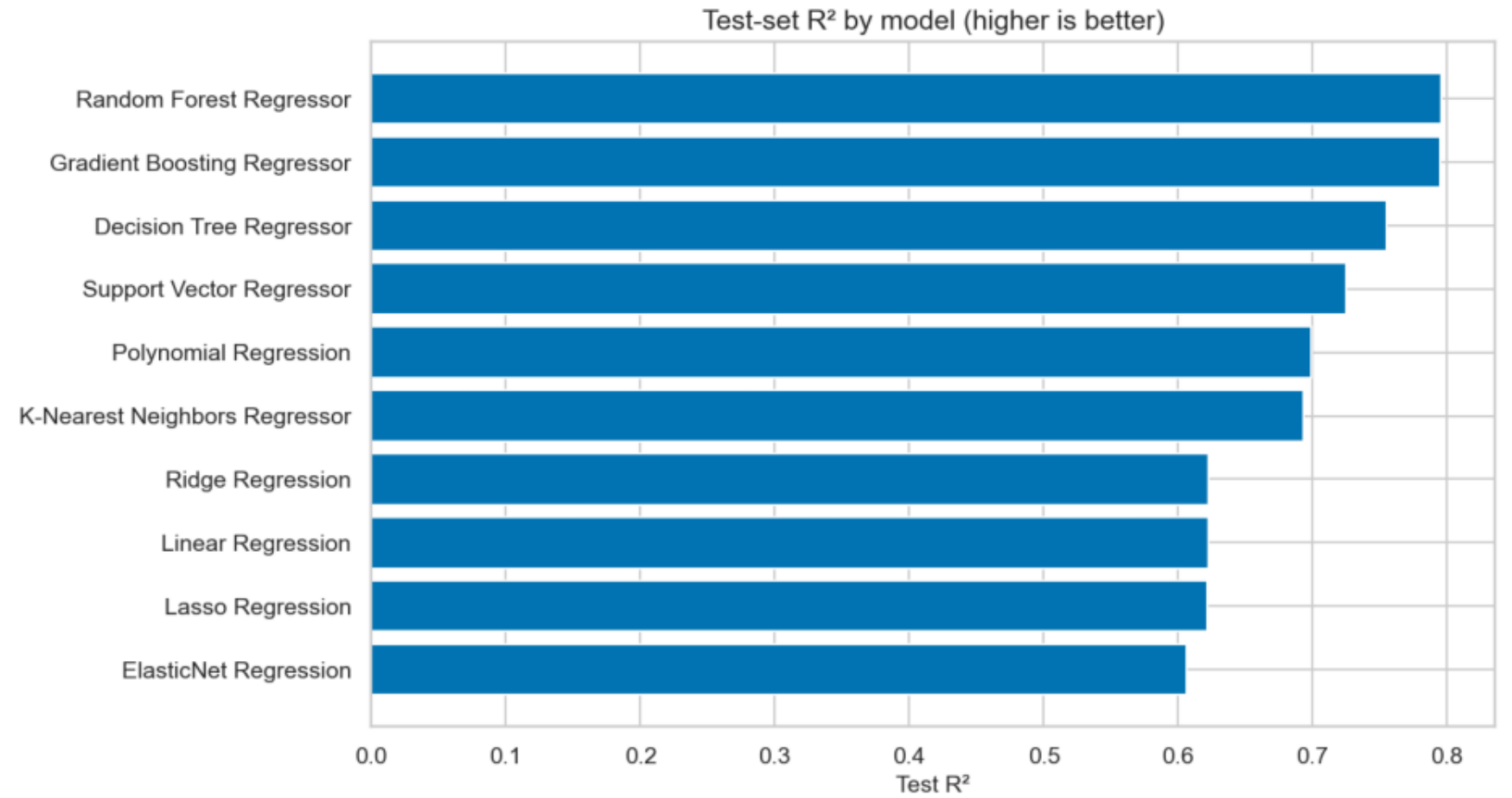


07 after cleaning distance vs duration



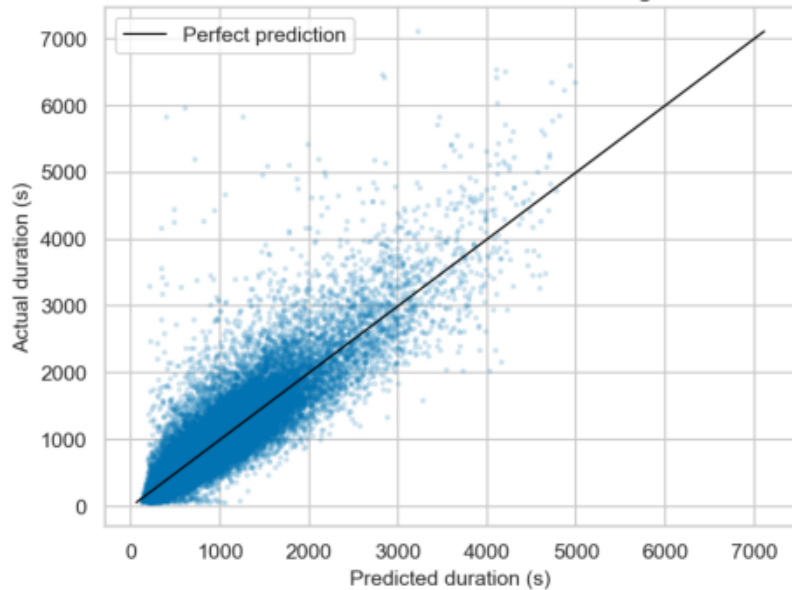


09 test set r^2 by model higher is better



10 predicted vs actual random forest regressor

Predicted vs actual — Random Forest Regressor



Residuals vs predicted — Random Forest Regressor

